

Examiner reports

Q1.

In part (a), the vast majority of students recognised the need to complete the square, with varying degrees of success. Most were able to remove a factor of 2, but less were able to

deal with the algebraic manipulation to achieve the required $\left(x - \frac{5}{4}\right)^2$. One third of all students went on to obtain the correct final result in the given form. Some students expanded and equated coefficients, but did not always put their result back in the required form.

In part (b), students did not always use inequalities as required. A variety of approaches were employed, including using the y coordinate of the vertex, forming a quadratic and using the discriminant and calculus.

Q2.

Part (a) was well attempted and most students gained some credit. Too many students lost marks for poorly drawn curves, which had the wrong curvature, lacked symmetry or were excessively wobbly.

The most frequent mark for part (b) was 2/4. The question discriminated well between students of varying abilities. While most students correctly used the discriminant > 0 for two real roots, many overlooked the fact that the roots were both positive, so only found half of the solution. Few students achieved full marks as insufficient justification/explanation was seen.

Q12.

- (a) (i) Many students were successful in completing the square, although some gave their answer as $(x - 3)^2 + 20$ instead of $(x - 3)^2 + 2$.
- (ii) Most students ignored the request in bold type and considered the discriminant of the quadratic. An algebraic approach was expected here, but few made much progress beyond writing $(x - 3)^2 = -2$. It was necessary to reason that no real square root of -2 exists and therefore the equation has no real solutions. Other acceptable reasoning was based on the fact that the minimum value of the expression $(x - 3)^2 + 2$ was equal to 2.
- (b) (i) The term “vertex” is becoming better understood and most students realised that they could write down its coordinates from the completed square form of the quadratic. A few used differentiation to find the stationary point, which was also acceptable.
- (ii) Those with the correct vertex usually drew a correct parabola with the value of the intercept on the y -axis clearly shown or stated.
- (iii) Many more students are now using the correct term “translation”, but many still used an incorrect word such as “move” or “shift” or also involved another transformation such as a “stretch”. The most common, but incorrect, vector

stated by students was $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and this suggests that they had not read the question carefully enough.

Q13.

- (a) Many students began by trying to complete the square for both the x terms and the y terms. These students, in an attempt to put $x = 0$, often simply wrote $(y - 2)^2 = 12$ and scored no marks. A surprisingly small number of students substituted $x = 0$ directly into the original equation; those who followed this method sometimes stopped at $y^2 - 4y = 12$, apparently being unable to solve the quadratic equation; others concluded that “ $y = 12$ or $y - 4 = 12$ ”, which is a bad error and not really expected at this level of mathematics. Those trying to solve $(y - 2)^2 = 16$ tended to miss the negative square root of 16.

Students who found that the radius was 5 sometimes used a diagram with a right angled triangle, two of the sides being 3 and 5. This enabled them to find the y -intercepts using the y -coordinate of C , by writing $y = 2 \pm 4$.

- (b) In their working for the previous part, many had already established the equation of the circle in the form $(x + 3)^2 + (y - 2)^2 = 25$ and could write down the value of the radius. However, the correct method was in doubt when 25^2 , or $\sqrt{25}$, instead of 25, appeared on the right hand side of the student's equation, even though in each of these cases the radius was then stated as being 5.

A significant number of students, despite the coordinates of the centre being given in the question, wrote down an incorrect circle equation such as $(x - 3)^2 + (y + 2)^2 = 25$ and once again concluded that the radius was equal to 5. Students cannot expect to score full marks for any answer that follows incorrect working.

- (c) (i) Although most students recognised that they should use the distance formula, many added 25 and 9 to make 36 not 34. There were also many examples of poor notation, such as $5^2 + 3^2 = \sqrt{34}$. There were many arithmetic slips due to the negative coordinate of C . Those who used an incorrect formula for the length of CP scored no marks at all.
- (ii) Some realised that $(2, 2)$ was one of the points of contact and then found PQ very easily. Failing to recognise the position of the right angle often led to many students writing $\sqrt{34 + 25}$ for the length of PQ . Many, who perhaps believed that they were using Pythagoras correctly, wrote $\sqrt{34} - \sqrt{25} = \sqrt{9}$ and this scored no marks, even though students obtained the correct value for the length of PQ . The mention of “tangent” often triggered attempts to find equations rather than using the geometrical properties of the circle.

Q14.

- (a) Students must be aware that, when they are asked to complete a proof, each stage must be clearly and accurately shown. This was not always the case, and careless sign errors and the omission of “ $= 0$ ” were penalised.

- (b) (i) A very common error here, in finding the discriminant, $b^2 - 4ac$, was where students explicitly wrote " $b = 2x + 1$ " instead of " $b = -(2x + 1)$ ". Those who did have the correct value of b sometimes wrote b^2 as $-(2x + 1)^2$. Unfortunately, many students did not write the condition for distinct real roots until the very last line which was insufficient, since this was the given answer.

The better solutions began by quoting the condition that two distinct real roots occur when $b^2 - 4ac > 0$ or the appropriate inequality with the discriminant expressed in terms of k .

- (ii) It was pleasing to see some completely correct solutions here. Those who chose to factorise the equation, in order to find the critical values, were usually more successful than those using the quadratic equation formula, as

$\sqrt{20^2 - 16 \times 9}$ defeated many without the use of a calculator. Students are advised to draw a sketch or sign diagram in order to solve a quadratic

inequality. A few thought that the two inequalities, $k < \frac{1}{2}$, $k > \frac{9}{2}$ should be combined and so spoiled their correct solution by writing something like $\frac{1}{2} > k > \frac{9}{2}$ as their final answer, and this was penalised.

Q15.

The majority of candidates failed to answer this question well.

- (a) (i) Many candidates were unsuccessful in completing the square as they did not realise the need to take out a factor of 2 first, so $2(x + 3)^2 - 4$ was a common incorrect answer. Many who made the correct approach did not handle the

factor of 2 correctly and the expression $2\left(x + \frac{3}{2}\right)^2 + \frac{1}{4}$ was seen quite often. Others made sign errors so another common incorrect answer was

$$2\left(x + \frac{3}{2}\right)^2 + \frac{11}{4}.$$

- (ii) Again there were few correct answers. Many candidates gave the value of x

for which the expression was a minimum, namely $x = -\frac{3}{2}$. Those who had completed the square correctly often offered a pair of coordinates, namely

$$\left(-\frac{3}{2}, \frac{1}{2}\right),$$

as their answer instead of the minimum value of $\frac{1}{2}$.

- (b) (i) Many candidates had no idea how to find the distance between the two points and attempts such as the sum of the products of both x coordinates and both y coordinates were seen. This may have been due to a lack of understanding of AB^2 . Some also interpreted it as $A \times B^2$. Those who began by simplifying $(3x + 9 - 5)^2$, rather than producing an expression with up to nine terms, usually went on to complete the proof successfully. A few made sign errors when subtracting the y coordinates, which led to errors. There was some poor

squaring such as $(x + 3)^2 = x^2 + 9$, though this did not occur often. It was good to see that almost all included " $AB^2 =$ " in their solution.

- (ii) There were very few correct solutions here. Hardly any candidates heeded the instruction of the question to use 5 times their value from part (a)(ii); those who attempted this part often substituted a value for x (usually in the form of a fraction) into $5(2x^2 + 6x + 5)$ and the simplification defeated them. Even some

of the most able candidates were unable to change $\sqrt{\frac{5}{2}}$ into the form $\frac{1}{2}\sqrt{n}$.

Q16.

- (a) Provided candidates began by writing $b^2 - 4ac \geq 0$, and did not write the coefficient of x incorrectly by writing $b = k - 2$ instead of $-(k - 2)$, then full marks were often earned. Some wrote $b^2 = -(k - 2)^2$, however, and then conveniently "lost" the minus sign. There were not many sign errors, possibly because the required inequality was given in the question.
- (b) There were several completely correct solutions. Those who chose to factorise the equation to find the critical values were usually more successful than those using the quadratic equation formula, as $\sqrt{48^2 - 4 \times 7 \times 80}$ defeated many without the use of a calculator. Candidates are advised to draw a sketch or sign diagram in order to solve a quadratic inequality. Some gave their final answer as $k \leq 4$, $k \geq \frac{20}{7}$ or $\frac{20}{7} < k < 4$, both of which were penalised by one mark.

Q17.

In this question, which tested logarithms, most students scored the mark in part (a) but not all could see its use at a later stage in part (b). Not unexpectedly, part (b) proved discriminating, with a majority of students only picking up one of the six marks, normally for writing $2\log(x + 7)$ as $\log(x + 7)^2$. The remaining students, who correctly applied a

further law of logarithms to reach the stage $\frac{(x+7)^2}{x+5} = 3$, either stopped at that point or went on to find the correct quadratic equation. In the latter case, most went on to solve the equation correctly but the required concluding statement 'only one value of x ' was not always given and so the final mark was lost. Some students who recognised what was being asked for frequently obtained the correct quadratic equation, showed that $b^2 - 4ac = 0$ and then stated 'only one value of x ' for full marks.

Q18.

Sketching a convincing tangent graph, even with graphical calculators available, seemed a considerable challenge for many. Three separate branches were needed with the correct intercepts on the x -axis and demonstration of the tendency near 90° and 270° . It was disappointing to sometimes see a sine or a cosine graph drawn instead. Many students correctly solved the trigonometric equation in part (a)(ii) although a minority of students incorrectly gave more than two solutions inside the given interval. The majority of students were able to recall and use the required two trigonometric identities in part (b)(i)

but not all were able to reach the printed answer without there being a sign error in their working. In the final part of this last question on the paper it was important that students recognised that they were required to use 'Hence'. and also that the equation to be solved was in terms of $3x$. Those students who just solved the equation in part (b)(i) without ever dividing its solutions by 3 or writing ' $\theta = 3x$ '. were heavily penalised. Those students who did link θ to $3x$ generally recognised and solved the quadratic in $\cos 3x$ but a significant minority then failed to find a third solution for x in the given interval.

Q19.

This was a very well answered question with a high proportion of students gaining full marks. A small number of students failed to round results to three significant figures or gave their answers in degrees. Some lost marks through sign errors when collecting the terms of the quadratic together prior to factorisation. The most fundamental error seen was where students tried to write everything in terms of $\tan x$ and occasionally the equation $8(1 + \tan x) - 2(1 + \tan^2 x) = \tan^2 x - 2$ was seen. Those who changed to sine and cosine at the start had to convert to an equation all in cosine to earn marks and some did not go far enough.

Q20.

In part (a), the factorisation was almost always correct. Perhaps because of a confusion between factors and roots, many solved an equation in this part to give $x = -2$ and $x = 6$. This was considered as further non-contradictory work and was not penalised.

In part (b), the most common score for the graph was 3 marks. A large number of candidates who had the correct x -intercepts drew a distorted parabola, which had a minimum at $(0, -12)$. Some were quite happy to mark the intercepts as $(-2, 0)$ and $(6, 0)$ but then drew a curve with these intercepts equidistant from the origin.

In part (c)(i), many candidates were familiar with the technique of completing the square, although quite a few gave their answer as $(x - 2)^2 - 8$ instead of $(x - 2)^2 - 16$.

In part (c)(ii), most candidates offered the coordinates of the minimum point of a curve, writing $(2, -16)$ as their answer, instead of giving **the minimum value of the expression**. Sadly, many did not return to part (b) to correct their graphs so as to illustrate this correct minimum point.

Part (d) was beyond the understanding of most candidates. A very common incorrect answer was $y = (x + 3)^2 + 2$. Some gained a mark for adding 2 to their quadratic (or for replacing y by $y - 2$) but many also replaced x by $x - 3$ instead of $x + 3$. Quite a number of those who used the correct approach did not actually write down **an equation**, forgetting to include " $y =$ ", and so lost the final mark.

Q21.

In part (a), most candidates were able to complete the square for at least one of the terms. A very common incorrect answer involved $(x - 7)^2$ instead of $(x + 7)^2$. Occasionally a negative answer was given for the square of the radius. Also, $\sqrt{5}$, $\sqrt{25}$ and ± 49 were often seen on the right-hand side of the equation.

In part (b), the coordinates of the centre usually followed through correctly from part (a). Often the correct radius was stated, even though part (a) was incorrect.

In part (c), the centre of the circle usually appeared in the correct quadrant for the candidate's coordinates. However, those who had the correct answers in part (b) did not always draw the correct circle touching the x -axis at -7 and not touching the y -axis.

In part (d)(i), several candidates earned the method mark but the most common error lay in the failure to square (kx) as k^2x^2 ; most wrote kx^2 and immediately lost the accuracy mark.

In part (d)(ii), candidates seemed to think it necessary to write " $= 0$ " only on the very last line. This does not fulfil the requirements of a proof. The first line should be that equal roots occur when $b^2 - 4ac = 0$ and the " $= 0$ " should continue throughout.

Other common errors occurred in the squaring of $2(k + 7)$, which often became $2(k + 7)^2$. Candidates should check each line of their proof instead of fudging one or more lines of working, as examiners are unlikely to be deceived by the miraculous appearance of the printed answer following incorrect working.

Many did not attempt part (d)(iii). The factorisation of the quadratic expression defeated most candidates; too many are reliant on the quadratic equation formula which often produces large numbers that are difficult to handle without a calculator. The incorrect factors $(4k - 3)(3k + 4)$ were seen as often as the correct ones. Candidates found this factorisation extremely hard and very few picked up these two final marks, but this is a topic that should not be beyond students at this level.

Q22.

In part (a)(i), the coefficient of x , being an odd number, caused problems for some when completing the square; most students could not square 1.5 without a calculator; those who

used fractions could not always simplify $5 - \frac{9}{4}$, and for those using decimals $-1.25 + 5 = 3.75$ was commonly seen.

In part (a)(ii), the equation of the line of symmetry was not usually stated correctly, with several writing down the equation of a curve rather than a straight line; the incorrect

equation $y = \frac{3}{2}$ was seen far too often, whereas others simply wrote down numbers such as $\frac{3}{2}$ or $\frac{11}{4}$; this topic was not well understood by many students.

In part (b)(i), most attempts earned the method mark for eliminating y and collecting like terms, but, surprisingly, not all were able to solve the resulting quadratic equation. A common incorrect solution to $x^2 - 4x = 0$ was $x = \pm 2$, whereas others obtained $y = x^2 - 4x$ from the two equations and made no progress.

Integration of polynomials is well drilled and part (b)(ii) earned full marks for weak and strong students alike. There were few errors seen but the most common involved the constant term; some kept the term as 5 and others omitted the final term possibly confusing differentiation with integration.

In part (b)(iii), most students realised the need to substitute appropriate limits but, without

a calculator, arithmetic errors were abundant; 4^3 was often evaluated as 48 and the

fractions and minus sign caused problems for many students who could not simplify $\frac{64}{3} - 24 + 20$.

Those who were unable to complete part (b)(i) often chose the value 5 as their upper limit, possibly since the coordinates of A were $(0, 5)$, but no credit was given for this. However, those who had the solution $x = 2$ from their earlier work were given a method mark for using this value as their upper limit. A large number of students presented the value of the definite integral as their final answer to the area of the shaded region instead of considering the difference of the area of the appropriate trapezium and the value of the integral.

Q23.

The sketch of the given exponential curve was generally well-done although some students did not display sufficient care in sketching the curve in the negative- x region. In part (b), the majority of students were awarded the mark for correctly eliminating one of the variables, but a significant number could not form a quadratic with any confidence. Those who recognised that 7^{2x} could be written as $(7^x)^2$ usually had no difficulty in proceeding to form and solve a correct quadratic equation in y and then to find the correct value for the y -coordinate of the point of intersection, although a small minority of these students did not give sufficient consideration to the rejection of the inappropriate negative solution. The usual errors seen are illustrated by the following: $7^x = 7^{2x} - 12 \Rightarrow 12 = 7^{2x} - 7^x \Rightarrow 12 = 7^x$ or $7^x = 7^{2x} - 12 \Rightarrow \log 7^x = \log 7^{2x} - \log 12$. There were, however, an encouraging number of completely correct solutions.

Q24.

In part (a) it was good to see almost everyone being able to differentiate correctly.

A few then went on to divide their expression for $\frac{dy}{dx}$ by 2 or 3 or 6, but this was not penalised at this stage.

In part (b) most candidates tackled the verification that P was a stationary point first and generally did so correctly, although some did not present accurate working, writing things such as $18 + 6 - 12 = 0$. Many stopped at this point, although a few tried listing

various values of x with the corresponding value of $\frac{dy}{dx}$, and occasionally discovered

the other value of x for which y was stationary. Many candidates equated $\frac{dy}{dx}$ to zero, but not all were then able to factorise the quadratic correctly. Some felt the need to use the quadratic equation formula and were not always successful in finding the correct solutions.

Part (c)(i) Those who had the correct expression for $\frac{dy}{dx}$ usually found the second

derivative correctly. Many ignored the request to find the value of $\frac{d^2y}{dx^2}$ when $x = -1$, but this was sometimes generously credited in part (c)(ii), where it was pleasing to see

most candidates being able to interpret the sign of the second derivative in order to describe the nature of the stationary point.

Q25.

In part (a)(i) Although the majority of candidates found the correct gradient, some lost

the $-$ sign and others wrote their answer as $-\frac{3}{2}x$. Those who used two correct points often made a sign error in simplifying their expression for the gradient.

In part (a)(ii) it was disconcerting to see so many candidates finding the equation of a line perpendicular to AB rather than the equation of a line parallel to AB . Even having the correct line equation did not mean that full marks were earned; quite a few found the value of x when $y = 0$ and others who correctly obtained the equation $3x + 2y + 8 = 0$ quoted the intercept as being at $y = -8$.

Part (b) was generally well done, although eliminating y and getting as far as $5x = -5$ did not always result in the correct solution; final answers of $x = -5$ or $x = 1$ were seen far too often.

In part (c) the relevant distance formula expression often had a sign error in at least one bracket or was equated to 5 instead of 25. However it was good to see a sizeable number of fully correct solutions. Some candidates used a diagram and a vector approach and wrote down the correct values of k .

Q26.

Part (a)(i) was quite often not attempted. Many candidates did not recognise the equation as that of a cubic and drew a parabola. Some drew the graph of $y = x^3$ with a stationary point of inflection but this scored no marks. The majority of candidates drew a cubic curve with a maximum point and a minimum point, but often these points were in the wrong place. Often those whose curve touched the x -axis at $x = 2$ did not draw their curve through the origin.

In (a)(ii) many candidates failed to score the mark here because of a variety of algebraic errors, often involving signs or a failure to write $= 0$ at the end of lines of working.

In (b)(i) the factor theorem was also well done but the necessary statement about $(x - 3)$ being a factor was occasionally missing. It was good to see only a few solutions using long division as they were given no credit for an incorrect method.

In part (b)(ii) the factorisation here was generally correct, and given in the required form. Many candidates were able to write down the quadratic factor by inspection.

In part (c) the more able candidates coped well with this part, establishing the value of the discriminant ($b^2 - 4ac$) for the quadratic factor and explaining that the value of -3 was negative and hence the quadratic equation had no real roots. Equally acceptable was an attempt to solve the equation and to comment about the negative value under the square root sign. The mark for identifying the real root as being 3 was earned by many but not by those who said that the root was $(x - 3)$ or other incorrect statements. Some weaker candidates wrongly tried to find the value of the "discriminant" using the coefficients of the cubic equation.

Q27.

In part (a)(i) the negative coefficient of x^2 confused many candidates. The correct answer was rarely seen. Those who chose to work initially with the negative of the given expression, even successfully finding $(x + 5)^2 - 29$, often wrote their final answer as $29 - (x - 5)^2$. Candidates need to practise completing the square for expressions of this type even though it has not previously been tested very often.

In (a)(ii) too many had no idea how the line of symmetry related to the previous part. Many candidates just wrote down the maximum point or the equation of the parabola.

In part (b)(i) although the majority of candidates realised that they should equate the two equations, numerous algebraic errors appeared before the final line and even this final equation was sometimes incorrect.

In part (b)(ii) even some able students were unable to cope with squaring $2(2k + 5)$ in order to find an expression for the discriminant. Again, poor algebra and fudging abounded in order to obtain the printed inequality. To earn all the marks it was essential to have stated the condition $b^2 - 4ac > 0$ early in their working.

In (b)(iii) Only correct factors or quadratic formula earned the first mark; candidates at this level should all be capable of this, particularly with 29 being a prime number. Unfortunately, even with correct factors, some lost the minus sign on one of their critical values. Many failed to draw the appropriate sketch or sign diagram for determining the correct solution, and a few lost the final mark by trying to combine the two correct inequalities thus producing an intransitive statement

such as $-\frac{29}{4} > k > -1$.

Q28.

There was a wide range of responses to this question and relatively few candidates gained full marks.

In part (a)(i) most candidates wrote down a correct form for $\tan 2x$ with only a few writing $\tan^2 x$ instead of $2\tan x$. Most multiplied by the denominator to obtain a cubic equation in $\tan x$. However, many divided throughout by $\tan x$ to get $\tan^2 x = 3$ and failed to demonstrate that $\tan x = 0$ is also a solution. Some used verification which did not warrant any marks.

In (a)(ii) very few candidates gained this mark. The solution of 120° was almost always omitted. Most wrote 0° and 180° too. This in itself was not penalised this time as these values were outside the range of x requested.

Part (b)(i) was answered very well, as candidates showed familiarity with double angle formulae and handled the algebra involved with confidence.

In part (b)(ii) the quadratic formula was used correctly by most candidates and most demonstrated where the $\sqrt{3}$ came from. The majority of candidates, though, just seemed to ignore the negative root and gave no explanation as to why. If they found only one root then they lost 2 marks. Some gave unsatisfactory explanations for the exclusion of the

negative root, such as $\sin x$ cannot be negative or that it was > 1 . Those who used completion of the square to solve the equation almost invariably just took the positive square root.

Q29.

In part (a) the coefficient of x , this time being an odd number, caused many problems in the completion of the square; most candidates could not square 2.5 without a calculator and

those who used fractions could not always simplify $7 - \frac{25}{4}$.

In (b)(i) most candidates realised that the vertex was the minimum point but failed to see the link with part (a) and hence were seldom able to write down the correct coordinates. A few were successful using differentiation, but even when the x -coordinate was correct, they often made arithmetic slips in finding the y -coordinate of the vertex.

In part (b)(ii) the equation of the line of symmetry was not usually correct with many writing down

the equation of a curve rather than a straight line; the incorrect wrong equation $y = -\frac{5}{2}$ was seen very often.

In (b)(iii) those with the correct minimum point were usually able to produce a correct sketch, although the value of the y -intercept was sometimes missing from good sketches. Some credit was given to candidates with an incorrect minimum point provided their graph was consistent with this minimum point.

In part (c) only the very able candidates who had answered part (a) correctly earned full marks here. The term **translation** was required but generally the wrong word was used or it was accompanied by another transformation such as a stretch (even though this transformation is not within the MPC1 specification). A very common (but incorrect) vector stated by candidates

was $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$, as once again many failed to realise the significance of part (a).

Q30.

In part (a) most candidates realised the need to find the value of $p(x)$ when $x = -1$. However, it was also necessary, after showing that $p(-1) = 0$, to write a statement that the zero value implied that $x + 1$ was a factor. It is encouraging to see an increasing number of candidates being more aware of this. Some evaluated $p(1)$ and scored no marks.

In (b)(i) those who used inspection were the most successful here; methods involving long division or equating coefficients usually contained algebraic errors.

In part (b)(ii) most candidates realised the need to consider the discriminant but several used the coefficients of the original cubic rather than those of the quadratic. Having shown that the value of the discriminant was negative it was necessary to state that the quadratic had no real roots and so the cubic had exactly one real root namely -1 .

Q31.

In part (a) most candidates found the correct values of a and b , but correct values for k were not so common. Some sloppiness was again evident with candidates failing to write squared outside the brackets or omitting the plus sign between the terms on the left hand side.

In (b) some used an approach based on Pythagoras, but the majority of candidates used an algebraic method, substituting $y = 0$ into their circle equation and solving the resulting quadratic equation in x . Although there were a number who made sign errors, many candidates produced the correct values of x . Nevertheless, common mistakes such as putting $x = 0$ or $(y + 8) = 0$, instead of $y = 0$, resulted in no marks being earned.

In part (c) many candidates assumed that the gradient of CA was 2.5 instead of finding the negative reciprocal and many who did have a correct equation were unable to produce an equation with integer coefficients as requested. A sketch might have helped some to have seen the correct configuration and perpendicularity.

In (d)(i) most candidates realised the need to substitute $y = 2x + 1$ into one or other form of their circle equation but algebraic errors abounded as they tried to produce the displayed quadratic equation. Those who expanded the circle equation first were generally more successful; others who did not have 100 on the right hand side of their circle equation failed to recognise this error when they failed to produce the printed answer for the equation.

Candidates should be strongly discouraged from starting each new line with an equals sign when producing an algebraic proof of this type that involves equations.

In (d)(ii) those who completed the square failed to score full marks if they simply wrote $x + 3 = \sqrt{11}$ before concluding that $x = -3 \pm \sqrt{11}$. Those using the quadratic equation formula

often struggled to simplify $\frac{-6 \pm \sqrt{44}}{2}$.

Q32.

Part (a) was well done.

In part (b) the majority recognised what was required and obtained the relevant quadratic even if their notation was sometimes a bit suspect. A few used a substitution for $\ln x$ which helped, but those who used x instead of, say Y , occasionally forgot to change back to earn the final marks. The occasional candidate changed $3\ln x$ to $\ln(x^3)$ which was not helpful, and after multiplication by $\ln x$ there were a few $(3\ln x)^2$ terms seen.

Q33.

A small number of candidates clearly did not recognise the modulus function at all.

In part (a)(i) most graphs were in the correct place and looked linear. However a few failed to extend into the first quadrant and many were lacking the values of either one or both intercepts and a few had the vertex labelled $-\frac{1}{3}$ or -3 .

In (a)(ii) although almost all graphs had the three requisite sections, quite a number had the outer parts linear or curving convexly instead of concavely. Many did not have all three intercepts correctly labelled.

In part (b)(i) the more able candidates were able to form two different quadratic equations and solve them, but many got only one pair of roots. However many formed wrong equations such as $3x + 3 = x^2 + 1$ as well and got extra roots. Having found correct roots some assumed that their negatives would also be roots, misunderstanding the modulus. Those who tried to square both sides usually made algebraic errors, and only a handful were able to establish correct roots from their quartic equation. Another misunderstanding about the modulus function caused some to discard the negative values that they had found.

In part (b)(ii) it was very common for the answers to be between their values rather than outside them. Only more able students produced the correct intervals.

Q34.

In part (a), the slightly unusual form of the initial quadratic caused problems to those who forgot to add 2 after multiplying out the brackets. The first term in the completion of the square was done successfully by most candidates, although the answer $(x - 4)^2 - 3$ was seen almost as often as the correct form. It was unfortunate that an error at this stage was seldom picked up by candidates when they went on to sketch the curve.

In part (b)(i), the sketch was usually of the correct shape but many drew a parabola with the vertex in the wrong quadrant, or the y -intercept was incorrect. The question did ask for the coordinates of the minimum point and this was not always stated by candidates.

In part (b)(ii), a few candidates immediately wrote down the correct equation for the tangent, whereas many felt the need to differentiate in order to find the gradient of the curve at the vertex. Many candidates seemed unaware that the vertex was actually the minimum point and that the tangent at the vertex would be parallel to the line $y = 0$.

In part (c), the more able candidates earned full marks. The term **translation** was required and, although there seemed to be an improvement in candidates' using the correct word, it was still common to see words such as "shift" or "move" being used instead. Others thought that simply writing down a vector was enough. A very commonly

stated, but incorrect, vector was $\begin{bmatrix} -4 \\ 1 \end{bmatrix}$.

Q35.

Part (a)(i) Candidates did not seem well drilled in completing the square when the coefficient of x^2 is not equal to 1. It was very rare to see a correct answer here although a few did realise that $p = 5$. Clearly further practice is required at this type of question.

Part (a)(ii) Only the more able candidates were able to reason sufficiently well using the result from part (a)(i) as well as providing a concluding statement. No credit was given for using the discriminant to show that the equation had no real roots since the wording of the question excluded this approach.

Part (b)(i) This kind of question has been set several times before and the usual errors were seen. Candidates should state the condition for real roots ($b^2 - 4ac \geq 0$) and find an

expression in terms of k for the discriminant using the correct inequality throughout. The inequality is then reversed when multiplying by a negative number. Again, many candidates would benefit from practising this technique, using brackets where appropriate to avoid algebraic errors.

Part (b)(ii) The factorisation of the quadratic was usually correct, but several candidates

wrote down one of the critical values as $\frac{1}{7}$. Most found critical values and either stopped or immediately tried to write down a solution without any working. Candidates are strongly advised to use a sign diagram or a sketch graph showing their critical values when solving a quadratic inequality.



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